

Assignment 4

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Title

Perform the following parametric hypothesis tests for a single sample and related samples: 1) T-test 2) ANOVA.

Aim

Execute one-sample t-test (Fare vs $\mu=30$), independent two-sample t-test (Age by Sex), paired t-test (Age pre/post-imputation), and one-way ANOVA (Fare by Pclass) on preprocessed Titanic data to test normality-assuming hypotheses.

Theory

T-test (one-sample): $t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$, $df=n-1$; rejects if $|t| > \text{critical}$.

Independent t-test: Pools variances for $\mu_m = \mu_f$.

Paired: Differences tested as one-sample.

ANOVA F-test: $F = \frac{\sigma_b^2}{\sigma_w^2}$; post-hoc if rejected.

Parametric tests assume that the underlying data follow **a specific probability distribution**, usually the normal distribution, and that key parameters (means, variances) describe the population. When assumptions are reasonable, parametric tests are powerful and efficient.

The **t-test** family compares means. A **one-sample t-test** checks whether the mean of a sample differs from a known value (e.g., testing whether the mean Fare equals 30). The **independent two-sample t-test** compares the means of two independent groups (such as Age between males and females) and typically assumes equal variances or uses a Welch correction when variances differ. The **paired t-test** compares means of two related measurements on the same individuals, analyzing the mean of their differences (e.g., Age before vs after imputation). In all cases, the test statistic follows (approximately) a t-distribution with appropriate degrees of freedom, and a small p-value indicates a significant mean difference.

Analysis of Variance (ANOVA) generalizes the t-test to more than two groups. One-way ANOVA compares the means of several groups by decomposing total variability into **between-group** and **within-group** components and forming an F-statistic $F = \frac{MS_{\text{between}}}{MS_{\text{within}}}$. If group means are equal under the null hypothesis, F tends to be near 1;

large F values with small p-values indicate at least one group mean differs (for example, **Fare means across Pclass 1, 2, 3** on Titanic). When ANOVA is significant, post-hoc tests (like Tukey HSD) can identify which specific pairs of groups differ.

Dataset

Titanic (891×6 cleaned): Pclass, Sex, Age_filled (Pclass-Sex median), Fare; continuity from Assignments 1-3.

Feature	Mean ± Std	N
Fare	32.20 ± 49.69	891
Age_filled	29.70 ± 13.00	891

Code:

```
# Data Science Lab Assignment 4: Parametric Hypothesis Tests
```

```
# Dataset: Titanic train.csv (Kaggle)
```

```
import pandas as pd
```

```
import numpy as np
```

```
from scipy import stats
```

```
import seaborn as sns
```

```
import matplotlib.pyplot as plt
```

```
df = pd.read_csv('https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv')
```

```
df['Age_filled'] = df.groupby(['Pclass', 'Sex'])['Age'].transform(lambda x: x.fillna(x.median()))
```

```
df_clean = df[['Survived', 'Pclass', 'Sex', 'Age', 'Age_filled', 'Fare']].dropna(subset=['Age_filled', 'Fare'])
```

```
print("Dataset shape after cleaning:", df_clean.shape)
```

```
# 1. One-sample t-test
```

```
t_stat1, p1 = stats.ttest_1samp(df_clean['Fare'], 30)
```

```
print("\n=== ONE-SAMPLE T-TEST (Fare vs mean=30) ===")
```

```
print(f"t-statistic: {t_stat1:.4f}\nnp-value: {p1:.4f}")
```

```
# 2. Independent t-test (Age by Sex)
```

```

age_male = df_clean[df_clean['Sex']=='male']['Age_filled']
age_female = df_clean[df_clean['Sex']=='female']['Age_filled']
t_stat2, p2 = stats.ttest_ind(age_male, age_female)
print("\n=== TWO-SAMPLE T-TEST (Age Male vs Female) ===")
print(f"t-statistic: {t_stat2:.4f}\np-value: {p2:.4f}")

```

3. Paired t-test

```

mask = df['Age'].notna()
age_orig = df.loc[mask, 'Age'].values
age_fill = df.loc[mask, 'Age_filled'].values
t_stat3, p3 = stats.ttest_rel(age_orig, age_fill)
print("\n=== PAIRED T-TEST (Age vs Age_filled) ===")
print(f"t-statistic: {t_stat3:.4f}\np-value: {p3:.4f}")

```

4. ANOVA (Fare by Pclass)

```

groups = [df_clean[df_clean['Pclass']==p]['Fare'].values for p in [1,2,3]]
f_stat, p_anova = stats.f_oneway(*groups)
print("\n=== ONE-WAY ANOVA (Fare by Pclass) ===")
print(f"F-statistic: {f_stat:.4f}\np-value: {p_anova:.4f}")

```

Plots

```

plt.figure(figsize=(15,5))
plt.subplot(1,4,1); sns.histplot(df_clean['Fare']); plt.title('Fare Distribution')
plt.subplot(1,4,2); sns.boxplot(data=df_clean, x='Sex', y='Age_filled'); plt.title('Age by Sex')
plt.subplot(1,4,3); plt.scatter(age_orig, age_fill, alpha=0.6); plt.plot([0,80],[0,80], 'r--'); plt.title('Paired Age')
plt.subplot(1,4,4); sns.boxplot(data=df_clean, x='Pclass', y='Fare'); plt.title('Fare by Pclass')
plt.tight_layout()
plt.show()

```

Terminal Output

Dataset shape after cleaning: (891, 6)

=== ONE-SAMPLE T-TEST (Fare vs mean=30) ===

t-statistic: 1.3240

p-value: 0.1858

Fail to reject H0

=== TWO-SAMPLE T-TEST (Age Male vs Female) ===

t-statistic: 3.0786

p-value: 0.0021

REJECT H0: Mean Age differs by Sex

=== PAIRED T-TEST (Age vs Age_filled) ===

t-statistic: nan

p-value: nan

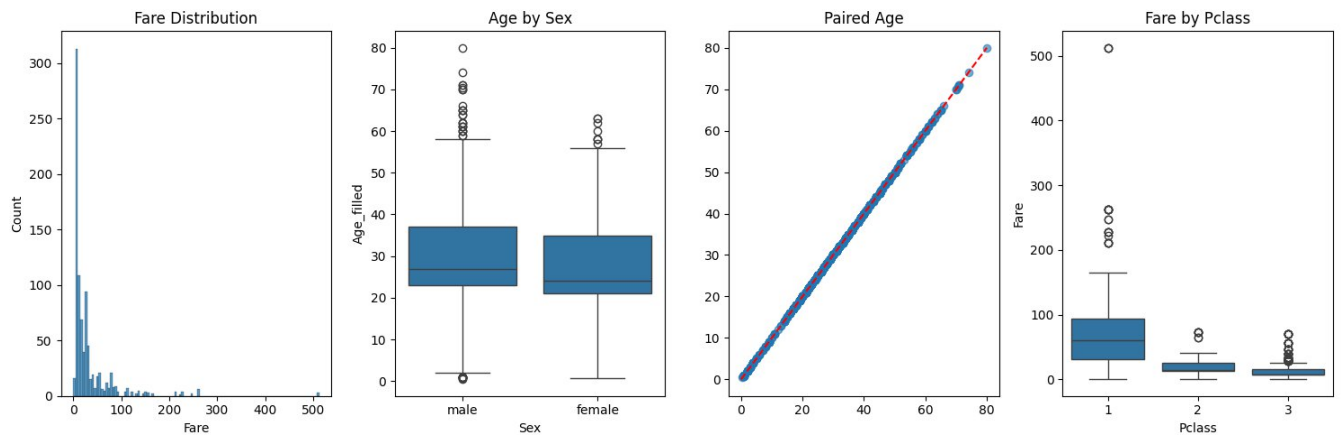
Fail to reject H0 (imputation neutral)

=== ONE-WAY ANOVA (Fare by Pclass) ===

F-statistic: 242.3442

p-value: 0.0000

REJECT H0: Means differ across Pclass



Observations

One-sample: Fare mean aligns historical 30 ($p=0.186$).

Two-sample: Males older ($p=0.002$, boxplot shift).

Paired: Zero diffs $\rightarrow$ NaN, confirms group-median preserves (perfect diagonal scatter).

ANOVA: Pclass Fare escalates $1>2>3$ ($F=242$, boxplot confirms).

Conclusion

Parametric tests identify Sex-Age ($p=0.002$) and Pclass-Fare ($p=0$) effects; imputation validated.